

# MOHAMMAD ALFAZ

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## PROFESSIONAL SUMMARY

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Full Stack and AI-focused Software Engineering student with hands-on experience building production-grade web applications using Python, Java, React, Next.js, and TypeScript. Skilled in designing REST APIs with FastAPI, modeling relational data in PostgreSQL, and applying Generative AI to solve real-world problems. Proven ability to ship end-to-end features independently across frontend, backend, and database layers, with a strong foundation in data structures, algorithms, and version control using Git. Seeking a Software Engineering or AI Engineering internship to apply strong problem-solving skills to high-impact engineering challenges.

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, C

**Frameworks & Libraries:** React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS

**Databases:** PostgreSQL, Supabase, MySQL, MongoDB

**Developer Tools:** Git, GitHub, VS Code, Postman, Vercel, Docker (basics), Linux

**AI & ML:** Generative AI, LLM API Integration, Prompt Engineering, NetworkX, OSMnx, Pandas, NumPy

**Core CS:** Data Structures & Algorithms, Object-Oriented Programming, REST API Design, Database Management Systems, Operating Systems, Graph Algorithms

## INTERNSHIPS

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**Java Development Intern** | Shadow Fox | Remote

2024

- Built a Bank Account Management System in core Java using object-oriented design principles, implementing classes for account creation, balance tracking, deposits, and withdrawals.
- Designed a modular package structure (*com.bank.core*) to separate business logic from application entry points, improving code readability and long-term maintainability.
- Developed an interactive Quiz application using JavaScript, HTML, and CSS, implementing dynamic question rendering, real-time score tracking, and instant UI updates without page reloads.
- Strengthened core Java fundamentals including OOP, exception handling, and collections by completing structured tasks under mentor guidance, translating requirements into working code independently.

**Full Stack Web Development Intern** | Prodigy InfoTech | Remote

2024

- Completed a four-project full stack development track, independently building and deploying responsive web applications using HTML, CSS, JavaScript, and modern frontend tooling.
- Engineered responsive, cross-browser UI components from scratch, applying semantic HTML and CSS best practices to ensure consistent rendering across devices and screen sizes.
- Implemented client-side interactivity using vanilla JavaScript, including DOM manipulation, event handling, and form validation, without relying on third-party UI libraries.
- Managed version control and project delivery using Git and GitHub, maintaining four separate repositories with clean commit history for each completed milestone.

## PROJECTS

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**VTU Buddy** | Next.js, React, TypeScript, PostgreSQL, Supabase

[github.com/alfaz1824/vtu-buddy](https://github.com/alfaz1824/vtu-buddy)

- Architected an AI-powered academic platform for VTU engineering students using Next.js and TypeScript, structuring a scalable App Router codebase for fast, SEO-friendly page loads.
- Designed a PostgreSQL schema on Supabase to manage previous year question papers, semester-wise study materials, and student resource metadata with efficient relational queries.
- Implemented secure user authentication and session management via Supabase Auth, enabling personalized access to saved resources and academic content.

- Built a fully responsive UI with reusable React components, ensuring seamless usability across mobile, tablet, and desktop breakpoints.
- Integrated backend APIs to dynamically fetch and render study materials and past papers, reducing manual content lookup for students searching scattered academic resources.

**EV Route Optimiser** | *Python, NetworkX, OSMnx, Graph Algorithms* [github.com/alfaz1824/ev\\_optimiser](https://github.com/alfaz1824/ev_optimiser)

- Developed a Python-based route optimization engine for electric vehicles, modeling real-world road networks as weighted graphs using OSMnx and NetworkX.
- Implemented shortest-path and energy-aware routing algorithms using Dijkstra-based pathfinding to compute energy-efficient routes, balancing distance against estimated energy consumption.
- Built a local caching layer to store processed map graph data, reducing redundant OSM data fetches and improving route computation speed on repeated queries.

**JeevaSetu AI** | *JavaScript, Node.js, Express, AI Integration* [github.com/alfaz1824/JeevaSetu-AI](https://github.com/alfaz1824/JeevaSetu-AI)

- Built a full stack AI-powered platform with a decoupled client-server architecture, separating a JavaScript frontend from an Express.js backend for independent scalability.
- Designed and exposed RESTful API endpoints to route data between the frontend interface and backend AI processing logic, delivering intelligent, context-aware responses to user input.
- Structured the codebase into clearly separated client and server modules, enabling independent development, testing, and future deployment of each layer.

**NirbhayaPath** | *React, Vite, Node.js, Mapping APIs* [github.com/alfaz1824/NirbhayaPath](https://github.com/alfaz1824/NirbhayaPath)

- Built NirbhayaPath, an AI-driven safety navigation platform designed to help users identify and travel safer routes in real time, using React with Vite for a fast, optimized frontend.
- Engineered a dedicated backend service (saferoute-server) to handle route computation and safety-related logic, decoupled from the client for independent scaling.
- Implemented interactive mapping functionality and designed the system around real-time, risk-aware routing concepts to help users make informed navigation decisions.

## EDUCATION

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**Bachelor of Engineering, Information Science & Engineering**

Visvesvaraya Technological University (VTU)

## CERTIFICATIONS / COURSES

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- IoT and Its Applications
- Gen AI Mastermind by Outskill
- Advanced Java
- Python Libraries for Data Science
- Microsoft Power BI

## HACKATHONS & ACHIEVEMENTS

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- Won the Future Innovation Excellence Award at Aavishkar Pravah 2.0, a national-level hackathon organized by Brainovisions in collaboration with HKBKCE.
- Won Best Project Award and a cash prize of Rs 5,000 in Full Stack Face Off, a web development competition conducted by HKBKCE.
- Secured Runner-up position in the Hackathon at Tech Takshak, a national-level tech fest organized by KNSIT.
- Won Best Student with a Vision Award in a 24-hour Web Development Hackathon organized by HKBKCE.